

Harry Greateorex, PhD

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TECHNICAL SKILLS

- **Programming & Tools:** Python (NumPy, pandas, SciPy, scikit-learn, PyTorch, Matplotlib) — primary language, 5+ years; SQL; IDL; Bash; Git; L^AT_EX; Excel and Microsoft Office; Linux/UNIX and high-performance computing (HPC) environments.
- **Data Analysis & Machine Learning:** Statistical analysis, data cleaning and validation, feature engineering, supervised classification with neural networks, principal component analysis (PCA), time-series analysis, data visualisation.
- **Data Pipelines & Automation:** End-to-end pipeline design; automated ingestion, cleaning, feature extraction, and classification; data reduction; pipeline and data-efficiency management; multi-terabyte datasets; remote compute, server, and database access.
- **Data Coordination:** Multi-source data alignment; data calibration; operations management.

ADDITIONAL SKILLS

- **Research & Project Management:** Project scoping and delivery; competitive funding acquisition; compute-time and resource allocation; supervision of junior researchers.
- **AI Tools:** Proficient with a range of AI models; experienced with Claude Code and Cowork; usage management, token efficiency, and quality control.
- **Communication & Dissemination:** Technical writing — peer-reviewed publications and code documentation; referee for scientific journals; reporting results to senior advisors and project leaders; conference talks, invited seminars, broadcast media, and public outreach.
- **Leadership:** Supervised a research intern and MSci students; chaired a session at an international scientific meeting; managed an amateur football team, including finances, sponsorship, and competition entries.

PROFILE

Research scientist working in applied computational astrophysics seeking to transition from an academic environment into a fast-paced, high-skilled, technical role. I have over five years of experience in independent and collaborative quantitative research, involving project conception, funding acquisition, data analysis, results delivery, and defence of findings. My work is focussed on the analysis of large, multi-source satellite datasets, statistical modelling, and machine-learning classification, with results published in peer-reviewed journals and presented across a spectrum of audiences, from NASA mission teams to international conferences to public outreach talks. I am now looking to transition into an industry where I can utilise my analytical skill set to provide science-quality solutions to real-world commercial problems.

EXPERIENCE

Postdoctoral Research Fellow April 2025 – Present
Astrophysics Research Centre, Queen's University Belfast Belfast, UK

- Led the largest statistical study of its kind in solar physics, analysing 15 years of NASA satellite data covering several thousand events. I designed and ran the full Python pipeline — ingestion, cleaning, feature extraction, and classification — and published the results as first author in June 2026.
- Built supervised machine-learning models in PyTorch to detect weak signals in noisy time-series data, applying class weighting and related techniques to work against a limited sample of labelled events.
- Ran GPU-enabled workloads in high-performance and remote computing environments, managing compute-time allocation and pipeline efficiency.
- Serve as a Chief Observer for the international Max Millennium programme since 2023, combining multiple live data feeds into a daily forecast issued to a global user community.
- Contribute to international instrument and mission development as a science-team member for the JAXA Solar-C/SoSpIM instrument and a contributor to the ESA FlaMEX mission proposal (shortlisted), working alongside teams across the US, Europe, and Japan.
- Conceived and delivered a team research project from conceptualisation through to publication: scoped the project, recruited and supervised a research intern, and managed the work to a peer-reviewed result; have mentored MSci students since 2022.
- Presented research to international audiences at conferences, invited seminars, and public outreach talks.

Visiting Scientist Jan – May 2023
NASA Goddard Space Flight Center Maryland, USA

- Secured competitive funding for overseas fieldwork, conducting a collaborative research project with mission scientists on the NASA Goddard campus.

Sales Assistant 2015 – 2020
Millets Derby, UK

- Met individual and team sales targets over five years in a KPI-driven store, working part-time alongside full-time study.

EDUCATION

PhD in Astrophysics

2021 – 2025

Queen's University Belfast

Thesis: *Observational Analysis of Lyman-alpha Emission from Solar Flares* — an extensive multi-instrument, multi-mission analysis of the solar plasma environment, particle properties, and terrestrial impact of the Sun's most powerful emission.

MSci in Physics (First Class Honours)

2017 – 2021

Queen's University Belfast

Thesis: *Computational Modelling of the Chromospheric Response to Solar Flares* — used 1D and 3D models of solar plasma properties to simulate particle interactions following a solar flare.

ADDITIONAL PROJECTS

- **SolarQuick** — a web application that consolidates the quicklook data tools used across the solar physics community into a single interface, built to address a community-wide gap in consolidated tooling. Developed using AI-assisted coding (Claude Code).
- **pydesat-aia** — an open-source Python tool for correcting saturated solar images, translated from earlier reference implementations in IDL and released with full documentation and usage notes.

MEDIA APPEARANCES

Solar Eclipse, Queen's University Belfast — August 2026. Broadcast and press interviews with BBC News NI, UTV (*On The Ground*, airing October 2026), Belfast Live / Daily Mirror, and The Irish News. Campaign generated 1,247 items of coverage worldwide (82.8% national, 13.4% international) with an advertising value equivalent of over £3.4 million.

PUBLICATION HIGHLIGHTS

Greatorex, H. J., et al., “The EUV Late-Phase: Statistical Results from 15 Years of Solar Dynamics Observatory Observations”, *Solar Physics*, 301, 6 (2026). DOI: 10.1007/s11207-026-02673-7

Greatorex, H. J., et al., “On the Instrumental Discrepancies in Lyman-alpha Observations of Solar Flares”, *Solar Physics*, 299, 162 (2024). DOI: 10.1007/s11207-024-02407-7

Greatorex, H. J., et al., “Observational Analysis of Lyman-alpha Emission in Equivalent Magnitude Solar Flares”, *The Astrophysical Journal*, 954, 120 (2023). DOI: 10.3847/1538-4357/acea7f

REFERENCES

References available on request.